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$\Rightarrow$ Laplace Transform:

Let f be a function defined for $t \geq 0$ then the integral:

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

is said to be the Laplace Transform of f , provided that the integral converges (means finite answer)

Q. Evaluate $\mathcal{L}\{1\}$

$$\text{Sol. } \mathcal{L}\{1\} = \int_0^{\infty} e^{-st} (1) dt$$

$$= \frac{e^{-st}}{-s} \Big|_0^{\infty} \Rightarrow = -\frac{1}{s} \left(e^{-st} \right) \Big|_0^{\infty}$$

$$= -\frac{1}{s} (e^{-\infty} - e^0) \Rightarrow \frac{1}{s} \quad \& \quad s > 0 \text{ Ans.}$$

$$\because e^{-\infty} = 0$$

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Q: Evaluate $\mathcal{L}\{t\}$

Sol: $\mathcal{L}\{t\} = \int_0^{\infty} e^{-st} t \, dt$ (By parts)

$$= \left[t \frac{e^{-st}}{-s} \right]_0^{\infty} - \int_0^{\infty} \frac{e^{-st}}{-s} (1) \, dt$$

$$= \lim_{t \rightarrow \infty} \left[\frac{t e^{-st}}{-s} \right] - \lim_{t \rightarrow \infty} \left[\frac{t e^{-st}}{-s} \right]$$

$$= \lim_{t \rightarrow \infty} \left(\frac{t}{-s e^{st}} \right) \Rightarrow = \frac{\infty}{\infty} \quad (\text{Apply L-Hôpital rule})$$

$$= \lim_{t \rightarrow \infty} \frac{1}{-s^2 e^{st}} \Rightarrow \frac{1}{\infty} \Rightarrow 0$$

Now the other half:

$$= 0 + \frac{1}{s} \int_0^{\infty} e^{-st} (1) \, dt \Rightarrow = \frac{1}{s} \cdot \frac{1}{s}$$

From previous

$$= 1/s^2 \quad \& \quad s > 0$$

Q: Evaluate $\mathcal{L}\{t^2\}$

Sol: $\mathcal{L}\{t^2\} = \int_0^{\infty} e^{-st} t^2 \, dt$

(From above ques)

$$= \left[t^2 \frac{e^{-st}}{-s} \right]_0^{\infty} - \int_0^{\infty} \frac{e^{-st}}{-s} (2t) \, dt$$

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$$= 0 + \frac{2}{s} \int_0^{\infty} e^{-st} t \, dt$$

$$= \frac{2}{s} \cdot \frac{1}{s^2} \Rightarrow = \frac{2}{s^3} \quad \& \quad s > 0 \quad \text{Ans.}$$

Q: Evaluate $\mathcal{L}\{t^3\}$

Sol: $\mathcal{L}\{t^3\} = \int_0^{\infty} e^{-st} t^3 \, dt$

$$= \left. \frac{t^3 e^{-st}}{-s} \right|_0^{\infty} - \int_0^{\infty} \frac{e^{-st}}{-s} (3t^2) \, dt$$

$$= \frac{3}{s} \int_0^{\infty} e^{-st} t^2 \, dt \Rightarrow = \frac{3 \cdot 2}{s^3}$$

$$= 6/s^4 \quad \text{OR} \quad \Rightarrow = 3!/s^4 \quad \& \quad s > 0$$

Q: Prove that $\mathcal{L}\{t^4\} = 4!/s^5 \quad \& \quad s > 0$
↳ Assignment 5.

$$\Rightarrow \mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, \quad s > 0$$

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 $\Rightarrow$ Gamma Functions.

$$\begin{aligned}
 \Gamma(n) &= \int_0^{\infty} e^{-t} t^{n-1} dt \\
 &= \frac{e^{-t} t^n}{+1n} \Big|_0^{\infty} - \int_0^{\infty} \frac{-e^{-t}}{n+1} t^n dt \\
 &= 0 + \frac{1}{n} \int_0^{\infty} e^{-t} t^n dt \quad \text{--- ①}
 \end{aligned}$$

Now by def of Γ :

$$\Gamma(n) = \int_0^{\infty} e^{-t} t^{n-1} dt$$

Put $n = n+1$

$$\Gamma(n+1) = \int_0^{\infty} e^{-t} t^n dt$$

So eq ① becomes:

$$\Gamma(n) = \frac{1}{n} \Gamma(n+1)$$

$$\Rightarrow \Gamma(n) = n \Gamma(n-1)$$

$$\Gamma(n) = n(n-1) \Gamma(n-2)$$

$$= n(n-1)(n-2) \Gamma(n-3)$$

$$= n(n-1)(n-2) \dots 2 \cdot 1 \Gamma(1)$$

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Now $\mathcal{L}(1)$:

$$\mathcal{L}(1) = \int_0^{\infty} e^{-t} dt \Rightarrow \left. \frac{e^{-t}}{-1} \right|_0^{\infty}$$

$$= 1 \quad (\text{So eq (2) becomes})$$

$$= [n(n-1)(n-2) \dots 2 \cdot 1] (1)$$

$$= n!$$

$$\mathcal{L}(n) = \int_0^{\infty} e^{-t} t^{n-1} dt$$

$$\text{impl } \mathcal{L}(n+1) = \int_0^{\infty} e^{-t} t^{n+1} dt \Rightarrow \int_0^{\infty} e^{-t} t^n dt = n!$$

from above

$$\text{Q. } \mathcal{L}\{t^n\} = \int_0^{\infty} e^{-st} t^n dt$$

$$\text{Sol. Put } st = u \Rightarrow t = u/s$$

$$dt = \frac{1}{s} du$$

$$= \int_0^{\infty} e^{-u} \left(\frac{u}{s} \right)^n \cdot \frac{1}{s} du$$

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$$= \frac{1}{s^{n+1}} \int_0^{\infty} e^{-u} u^n du$$

$\hookrightarrow n!$ (Proved on prev pg)

$$= \frac{n!}{s^{n+1}} \quad \text{Ans.}$$

Q: Find Laplace Transform of $f(t) = e^{at} \sin 2t$

Sol: $\mathcal{L}\{e^{at}\} = \int_0^{\infty} e^{-st} e^{at} dt$

$$= \int_0^{\infty} e^{-(s-a)t} dt \Rightarrow = \frac{e^{-(s-a)t}}{-(s-a)} \Big|_0^{\infty}$$

$$= 0 - \frac{1}{-(s-a)} \Rightarrow = \frac{1}{s-a}, \quad s > a$$

impl.

Q: $f(t) = \sin 2t$

Sol: $\mathcal{L}\{\sin 2t\} = \int_0^{\infty} e^{-st} \sin 2t dt$

$\hookrightarrow$ By parts

$$= \frac{e^{-st}}{-s} \sin 2t \Big|_0^{\infty} - \int_0^{\infty} \frac{e^{-st}}{-s} \cos 2t \cdot 2 dt$$

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$$= \frac{e^{-st} \sin 2t}{-s} \Big|_0^\infty + \frac{2}{s} \int_0^\infty e^{-st} \cos 2t \, dt$$

$$= 0 + \frac{2}{s} \left[\frac{e^{-st} \cos 2t}{-s} \Big|_0^\infty - \int_0^\infty \frac{e^{-st}}{-s} (-\sin 2t) 2 \, dt \right]$$

$$= \frac{2}{s} \left[0 - \left(-\frac{1}{s} \right) \right] - \frac{2 \times 2}{s \cdot s} \int_0^\infty e^{-st} \sin 2t \, dt$$

$$\mathcal{L}\{\sin 2t\} + \frac{4}{s^2} \int_0^\infty e^{-st} \sin 2t \, dt = \frac{2}{s^2}$$

$$\mathcal{L}\{\sin 2t\} \left(1 + \frac{4}{s^2} \right) = \frac{2}{s^2}$$

$$\mathcal{L}\{\sin 2t\} \left(\frac{s^2 + 4}{s^2} \right) = \frac{2}{s^2}$$

$$\mathcal{L}\{\sin 2t\} = \frac{2}{s^2 + 4} \quad \text{Ans.}$$

⇒ Assignment 5

Q: $f(t) = \cos 2t$, $f(t) = e^{2t+2}$

Q: $f(t) = e^{2t+2}$

Sol $\mathcal{L}\{e^{2t+2}\} = \int_0^\infty e^{-st} e^{2t+2} \, dt$

$$= \int_0^\infty e^2 e^{(2-s)t} \, dt \Rightarrow = e^2 \int_0^\infty e^{-(s-2)t} \, dt$$

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$$= e^2 \left[\frac{-e^{-(s-2)t}}{-(s-2)} \right]_0^\infty \Rightarrow = \frac{e^2}{s-2}, \quad s-2 > 0$$

short cut

Q: Find Laplace transform

$$f(t) = \underline{\sin kt} \quad \cos kt \quad (\text{Assignment 5})$$

$$\text{Sol: } \mathcal{L}\{\sin kt\} = \frac{1}{2i} \left[\mathcal{L}\{e^{ikt}\} - \mathcal{L}\{e^{-ikt}\} \right]$$

$$\because \sin kt = \frac{e^{ikt} - e^{-ikt}}{2i}$$

$$\because \mathcal{L}\{e^{at}\} = \frac{1}{s-a}, \quad s-a > 0$$

$$\mathcal{L}\{\sin kt\} = \frac{1}{2i} \left[\frac{1}{s-ik} - \frac{1}{s+ik} \right] \Rightarrow = \frac{1}{2i} \left[\frac{s+ik - s+ik}{s^2+k^2} \right]$$

$$= \frac{1}{2i} \left(\frac{2ik}{s^2+k^2} \right) \Rightarrow = \frac{k}{s^2+k^2}, \quad s^2+k^2 > 0$$

Ans.

$$\because i^2 = -1$$

Q: $f(t) = \cosh kt \quad \sinh kt$ (Assignment

$$\text{Sol: } \mathcal{L}\{\cosh kt\} = \mathcal{L}\left\{ \frac{e^{kt} + e^{-kt}}{2} \right\}$$

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$$= \frac{1}{2} \mathcal{L} \{ e^{kt} \} + \frac{1}{2} \mathcal{L} \{ e^{-kt} \}$$

$$= \frac{1}{2} \left[\frac{1}{s-k} + \frac{1}{s+k} \right]$$

$$= \frac{1}{2} \left[\frac{s+k + s-k}{s^2 - k^2} \right]$$

$$= \frac{1}{2} \left(\frac{2s}{s^2 - k^2} \right) \Rightarrow = \frac{s}{s^2 - k^2}$$

Q: Compute Laplace Transform of following fns:

① $f(t) = \cos^2 \omega t$

② $f(t) = \sin(\omega t - \phi)$

① Sol. $\mathcal{L} \{ \cos^2 \omega t \} = \mathcal{L} \left\{ \frac{\cos 2\omega t + 1}{2} \right\}$

$$= \frac{1}{2} \mathcal{L} \{ \cos 2\omega t \} + \frac{1}{2} \mathcal{L} \{ 1 \}$$

$$= \frac{s^2 + 2\omega^2}{s(s^2 + 4\omega^2)} \quad \text{Ans.}$$

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Q. Sol. $\mathcal{L}\{\sin(\omega t - \phi)\}$

$$= \mathcal{L}\{\sin \omega t \cos \phi - \cos \omega t \sin \phi\}$$

$$= \cos \phi \int_0^{\infty} e^{-st} \sin \omega t \, dt - \sin \phi \int_0^{\infty} e^{-st} \cos \omega t \, dt$$

$$= \cos \phi \left(\frac{\omega}{s^2 + \omega^2} \right) - \sin \phi \left(\frac{s}{s^2 + \omega^2} \right) \text{ Ans.}$$

Q: $\mathcal{L}\{f(t)\}$ where $f(t) = \begin{cases} 0 & 0 \leq t < 3 \\ 2 & t \geq 3 \end{cases}$

Sol. $\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) \, dt$

$$= \int_0^3 e^{-st} f(t) \, dt + \int_3^{\infty} e^{-st} f(t) \, dt$$

$$= \int_0^3 e^{-st} \cdot 0 \, dt + \int_3^{\infty} e^{-st} \cdot 2 \, dt$$

$$= 2 \int_3^{\infty} e^{-st} \, dt \Rightarrow = 2 \left[\frac{e^{-st}}{-s} \right]_3^{\infty}$$

$$= 2 \left[0 - \left(\frac{e^{-3s}}{-s} \right) \right] \Rightarrow = \frac{2e^{-3s}}{s}, \quad s > 0$$

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→ Add 5 ques of Laplace from chap 14 in assign 5.

Q: $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$

⇒ Periodic functions: A periodic fn is any fn for which $f(t) = f(t+T)$ for all T . The smallest constant T that satisfies (1) is called period of the fn.

Q. Find periodic fn of $f(t) = \sin t$.
↳ $\sin(t + 2m\pi)$

Q. Find periodic fn of $f(t) = \cos 3t$
↳ Ans: $2\pi/3$

Sol: $\cos 3(t+T) = \cos 3t$
 $\cos(3t + 3T) = \cos 3t$
 $3T = 2\pi$

$T = 2\pi/3$ period.

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$$Q. \int_0^{\pi} \sin t \, dt$$

$$= 2 \quad \text{Ans.}$$

$$Q. \int_{2\pi}^{3\pi} \sin t \, dt$$

$$= 2 \quad \text{Ans.}$$

$$Q.: \text{ Prove: } \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

↳ By Fourier

$$\text{Sol. } I = \int_{-\infty}^{\infty} e^{-x^2} dx = \int_{-\infty}^{\infty} e^{-y^2} dy$$

$$I^2 = \int_{-\infty}^{\infty} e^{-x^2} dx \cdot \int_{-\infty}^{\infty} e^{-y^2} dy$$

$$I^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy$$

↳ In polar form

→ circle mein angle
0 se 2π tk jao
skta.

$$I^2 = \int_0^{\infty} e^{-r^2} r dr \int_0^{2\pi} d\theta$$

$$dx dy = r dr d\theta$$

OR:

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta$$

$$I^2 = -\frac{1}{2} \int_0^{2\pi} e^{-r^2} (-2) r dr \int_0^{2\pi} d\theta$$

$$= \frac{1}{2} e^{-r^2} \Big|_0^{\infty} \cdot \theta \Big|_0^{2\pi}$$

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$$I^2 = \pi$$

$$I = \sqrt{\pi}$$

Ans.